

PlacementPal Knowledge Base

This document serves as a compact knowledge base for PlacementPal, covering the core areas required for campus placements, internships, and software engineering interviews.

1. Aptitude Topics

Percentages, Profit & Loss, Ratio & Proportion, Time & Work, Pipes & Cisterns, Speed Time Distance, Permutations & Combinations, Probability, Number Systems, HCF & LCM, Averages, Simple & Compound Interest, Data Interpretation, Logical Reasoning, Puzzles, Seating Arrangement, Blood Relations, Direction Sense, Coding-Decoding, Syllogisms, Calendars, and Clocks.

2. Data Structures & Algorithms

Arrays, Strings, Hashing, Linked Lists, Stacks, Queues, Recursion, Binary Search, Sliding Window, Two Pointers, Prefix Sum, Trees, Binary Trees, BST, Heaps, Graphs, Dynamic Programming, Greedy Algorithms, Backtracking, Tries, Union Find, and Advanced Graph Algorithms.

3. Core Computer Science

DBMS: Normalization, SQL Queries, Indexing, Transactions, ACID. Operating Systems: Processes, Threads, Scheduling, Deadlocks, Memory Management. Computer Networks: OSI Model, TCP/IP, HTTP, DNS, Routing, Subnetting. OOP: Encapsulation, Inheritance, Polymorphism, Abstraction, SOLID Principles.

4. Development Knowledge

Frontend: HTML, CSS, JavaScript, React. Backend: Node.js, Express. Databases: MongoDB, PostgreSQL, MySQL. APIs: REST, Authentication, JWT. Version Control: Git, GitHub. Cloud: AWS Fundamentals.

5. Interview Preparation

Prepare project explanations, STAR-based behavioral answers, resume walkthrough, strengths and weaknesses, leadership examples, conflict resolution stories, and internship/project experiences.

6. Company Assessment Strategy

Service-based companies: Aptitude + Basic Coding + Communication. Product-based companies: DSA + CS Fundamentals + Projects. Focus on timed mock tests and interview simulations.

7. Revision Checklist

Solve 150+ DSA problems, revise SQL, DBMS, OS, CN, OOP, complete aptitude practice, prepare project demos, update resume and GitHub, and practice mock interviews.